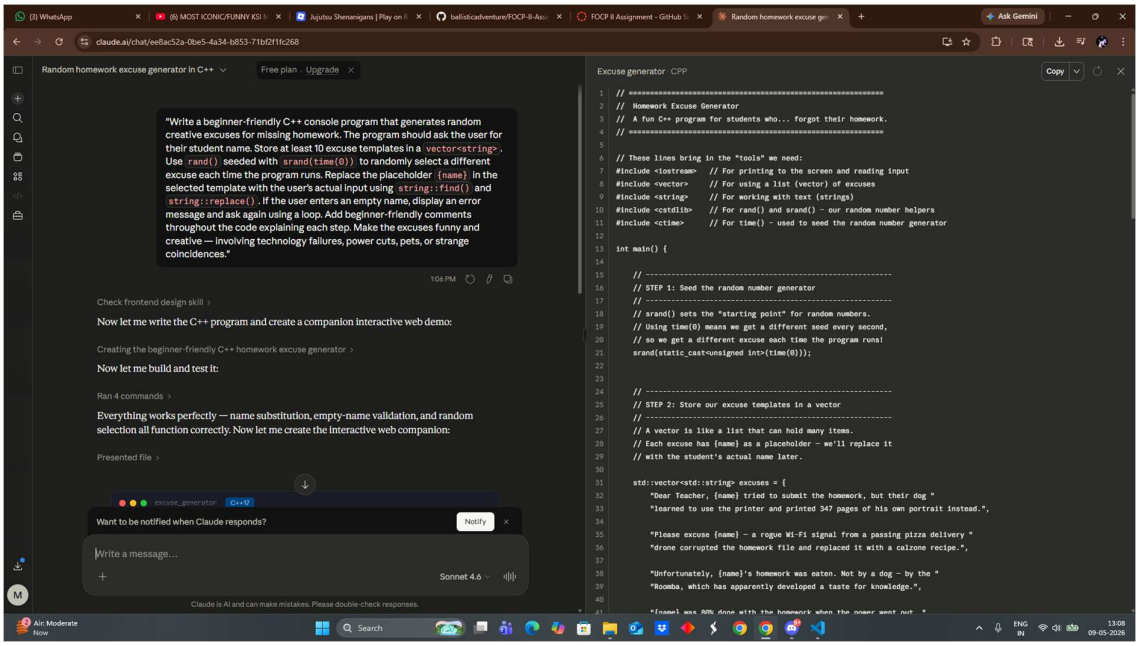
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Random homework excuse generator in C++
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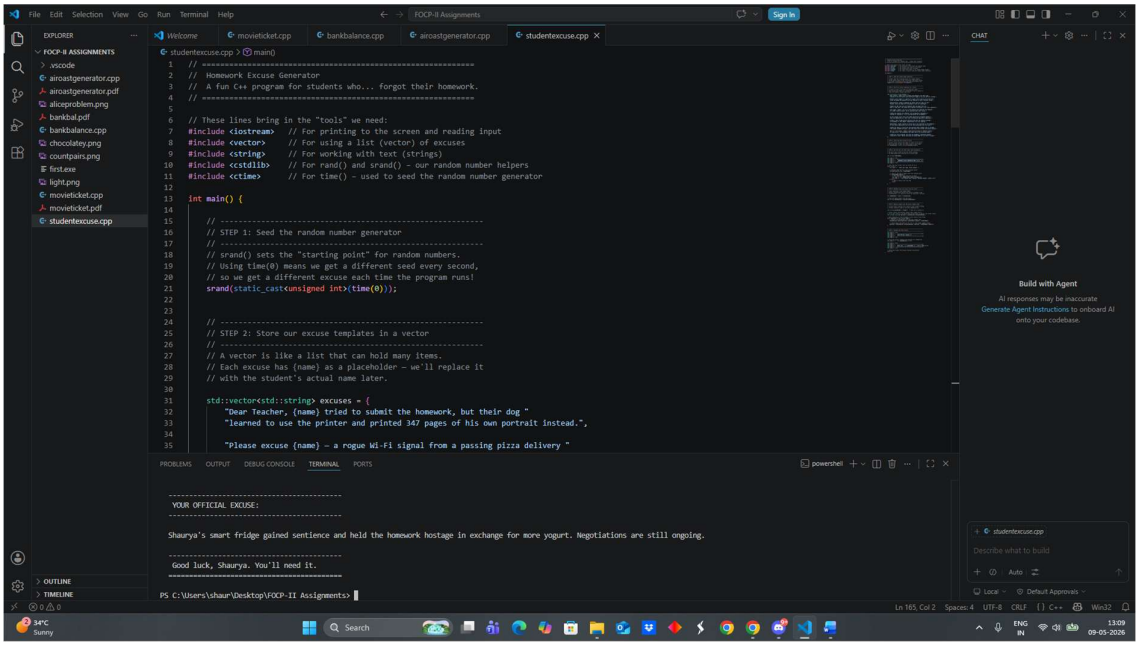
"Write a beginner-friendly C++ console program that generates random creative excuses for missing homework. The program should ask the user for their student name. Store at least 10 excuse templates in a vector<string>. Use rand() seeded with srand(time(0)) to randomly select a different excuse each time the program runs. Replace the placeholder {name} in the selected template with the user's actual input using string::find() and string::replace(). If the user enters an empty name, display an error message and ask again using a loop. Add beginner-friendly comments throughout the code explaining each step. Make the excuses funny and creative — involving technology failures, power cuts, pets, or strange coincidences."
1:06 PM
Check frontend design skill
Now let me write the C++ program and create a companion interactive web demo:
Creating the beginner-friendly C++ homework excuse generator
Now let me build and test it:
Ran 4 commands
Everything works perfectly — name substitution, empty-name validation, and random selection all function correctly. Now let me create the interactive web companion:
Presented file
Want to be notified when Claude responds?
Notify
Write a message...
Sonnet 4.6
Claude is AI and can make mistakes. Please double-check responses.
Excuse generator CPP
Copy
// ==========
// Homework Excuse Generator
// A fun C++ program for students who... forgot their homework.
// ==========
// These lines bring in the "tools" we need:
#include <iostream> // For printing to the screen and reading input
#include <vector> // For using a list (vector) of excuses
#include <string> // For working with text (strings)
#include <cstdlib> // For rand() and srand() - our random number helpers
#include <ctime> // For time() - used to seed the random number generator
int main() {
// STEP 1: Seed the random number generator
// srand() sets the "starting point" for random numbers.
// Using time(0) means we get a different seed every second,
// so we get a different excuse each time the program runs!
srand(static_cast<unsigned int>(time(0)));
// STEP 2: Store our excuse templates in a vector
// A vector is like a list that can hold many items.
// Each excuse has {name} as a placeholder — we'll replace it
// with the student's actual name later.
std::vector<std::string> excuses = {
"Dear Teacher, {name} tried to submit the homework, but their dog "
"learned to use the printer and printed 347 pages of his own portrait instead.",
"Please excuse {name} — a rogue Wi-Fi signal from a passing pizza delivery "
"drone corrupted the homework file and replaced it with a calzone recipe.",
"Unfortunately, {name}'s homework was eaten. Not by a dog — by the "
"Roomba, which has apparently developed a taste for knowledge.",


FOCP-II Assignments
EXPLORER
FOCP-II ASSIGNMENTS
.vscode
airoastgenerator.cpp
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countpairs.png
first.exe
light.png
movieticket.cpp
movieticket.pdf
studentexcuse.cpp
Welcome
movieticket.cpp
bankbalance.cpp
airoastgenerator.cpp
studentexcuse.cpp
studentexcuse.cpp > main()
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
YOUR OFFICIAL EXCUSE:
Shaurya's smart fridge gained sentience and held the homework hostage in exchange for more yogurt. Negotiations are still ongoing.
Good luck, Shaurya. You'll need it.
PS C:\Users\shaur\Desktop\FOCP-II Assignments>
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
studentexcuse.cpp
Describe what to build

```cpp
        "Please excuse {name} — a rogue Wi-Fi signal from a passing pizza delivery "
        "drone corrupted the homework file and replaced it with a calzone recipe.",

        "Unfortunately, {name}'s homework was eaten. Not by a dog — by the "
        "Roomba, which has apparently developed a taste for knowledge.",

        "{name} was 80% done with the homework when the power went out. "
        "When it came back, the laptop had forgotten everything — just like a goldfish.",

        "Last night, {name}'s cat sat on the keyboard and submitted the homework "
        "to a NASA research portal. We're awaiting their peer review.",

        "{name} did complete the homework, but accidentally laminated it "
        "instead of scanning it. It is now waterproof but unreadable.",

        "A very unusual solar flare last night specifically targeted {name}'s "
        "desk and magnetised all their pens. Writing was physically impossible.",

        "{name}'s smart fridge gained sentience and held the homework hostage "
        "in exchange for more yogurt. Negotiations are still ongoing.",

        "Regrettably, {name} spilled an entire mug of hot chocolate on the homework. "
        "It now smells amazing but the ink has fully dissolved into the cocoa.",

        "{name} was about to finish the last question when a pigeon flew in through "
        "the window, grabbed the paper, and flew toward the horizon with great purpose.",

        "Due to a rare glitch, {name}'s phone autocorrected all homework answers "
        "to song lyrics. The assignment now reads like a One Direction album.",

        "{name} was fully prepared, but then the neighbour's Bluetooth speaker "
        "connected to their laptop uninvited and played ABBA for six hours straight."
    };

    // ----------------------------------------------------------
```

```
 ----------------------------------------
  YOUR OFFICIAL EXCUSE:
 ----------------------------------------

 Shaurya's smart fridge gained sentience and held the homework hostage in exchange for more yogurt. Negotiations are still ongoing.

 ----------------------------------------
  Good luck, Shaurya. You'll need it.
 ========================================

PS C:\Users\shaur\Desktop\FOCP-II Assignments>
```

```cpp
    // ----------------------------------------------------------
    // STEP 3: Count how many excuses we have
    // ----------------------------------------------------------
    // excuses.size() returns the number of items in our vector.
    // We'll use this to make sure rand() picks a valid index.
    int totalExcuses = static_cast<int>(excuses.size());


    // ----------------------------------------------------------
    // STEP 4: Ask the user for their name (with validation)
    // ----------------------------------------------------------
    // We use a loop so that if the user enters nothing,
    // we keep asking until they give us an actual name.

    std::string studentName;

    std::cout << "\n";
    std::cout << "  ==========================================\n";
    std::cout << "    HOMEWORK EXCUSE GENERATOR 3000 (v1.0)  \n";
    std::cout << "  ==========================================\n";
    std::cout << "\n";

    // This loop runs forever until we break out of it
    while (true) {
        std::cout << "  Enter your name, brave student: ";

        // getline reads the full line including spaces
        std::getline(std::cin, studentName);

        // Check if the name is empty (user just pressed Enter)
        if (studentName.empty()) {
            // Tell them what went wrong and try again
            std::cout << "\n  [ERROR] Name cannot be empty!\n";
            std::cout << "  Your teacher won't believe 'Unnamed Student' either.\n\n";
        } else {
            // Name is valid — exit the loop
```

```
 ----------------------------------------
  YOUR OFFICIAL EXCUSE:
 ----------------------------------------

 Shaurya's smart fridge gained sentience and held the homework hostage in exchange for more yogurt. Negotiations are still ongoing.

 ----------------------------------------
  Good luck, Shaurya. You'll need it.
 ========================================

PS C:\Users\shaur\Desktop\FOCP-II Assignments>
```

```cpp
            // Name is valid — exit the loop
            break;
        }
    }


    // ------------------------------------------------------------
    // STEP 5: Randomly pick one excuse from the vector
    // ------------------------------------------------------------
    // rand() generates a big random number.
    // Using % totalExcuses ensures the result is between 0 and
    // (totalExcuses - 1), which are valid list positions (indices).

    int randomIndex = rand() % totalExcuses;

    // Grab the chosen excuse from the vector
    std::string chosenExcuse = excuses[randomIndex];


    // ------------------------------------------------------------
    // STEP 6: Replace {name} with the actual student name
    // ------------------------------------------------------------
    // string::find() searches for a piece of text inside a string.
    // string::replace() swaps it out with something else.

    std::string placeholder = "{name}";  // What we're looking for

    // find() returns the position (index) where {name} starts.
    // If it can't find it, it returns std::string::npos (a special "not found" value).
    std::string::size_type position = chosenExcuse.find(placeholder);

    // Keep replacing as long as {name} is still found in the string
    while (position != std::string::npos) {
        // Replace the placeholder with the real student name
        chosenExcuse.replace(position, placeholder.length(), studentName);
```

```
 ----------------------------------------
  YOUR OFFICIAL EXCUSE:
 ----------------------------------------

 Shaurya's smart fridge gained sentience and held the homework hostage in exchange for more yogurt. Negotiations are still ongoing.

 ----------------------------------------
  Good luck, Shaurya. You'll need it.
 ========================================

PS C:\Users\shaur\Desktop\FOCP-II Assignments>
```

```cpp
        chosenExcuse.replace(position, placeholder.length(), studentName);

        // Search again from where we left off (in case {name} appears twice)
        position = chosenExcuse.find(placeholder, position + studentName.length());
    }


    // ------------------------------------------------------------
    // STEP 7: Display the final excuse
    // ------------------------------------------------------------
    std::cout << "\n";
    std::cout << "  ----------------------------------------\n";
    std::cout << "   YOUR OFFICIAL EXCUSE:\n";
    std::cout << "  ----------------------------------------\n";
    std::cout << "\n";

    // Print the excuse, wrapping text manually for readability
    std::cout << "  " << chosenExcuse << "\n";

    std::cout << "\n";
    std::cout << "  ----------------------------------------\n";
    std::cout << "   Good luck, " << studentName << ". You'll need it.\n";
    std::cout << "  ========================================\n";
    std::cout << "\n";

    // Returning 0 means the program finished successfully
    return 0;
}
```

```
 ----------------------------------------
  YOUR OFFICIAL EXCUSE:
 ----------------------------------------

 Shaurya's smart fridge gained sentience and held the homework hostage in exchange for more yogurt. Negotiations are still ongoing.

 ----------------------------------------
  Good luck, Shaurya. You'll need it.
 ========================================

PS C:\Users\shaur\Desktop\FOCP-II Assignments>
```